

ELLIOT HAWKINS

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EDUCATION

Master of Science in Electrical Engineering and Computer Science; Electrical Engineering Emphasis Fowler School of Engineering, Chapman University, Orange, CA	May 2027
Bachelor of Science in Electrical Engineering; Minor: Music Technology Fowler School of Engineering, Chapman University, Orange, CA	May 2026 3.92 GPA

TECHNICAL SKILLS

Circuit Analysis | PCB Design and Fabrication | Python and C/C++ | LTSpice | Soldering THT & SMD | Laser Cutting
3D Printing (FDM, resin, SLS) | Signal Processing | Multimeter & DMM | Oscilloscopes | Bench power supplies | Arduino
3D CAD | Project Management | MatLab | System Verilog | Waveform Analysis | Adobe Suite | CNC milling | Prototyping

RELEVANT COURSES

Electronics and Circuits II | Control Systems | Electromagnetics I | Digital Logic Design II | Systems Programming
Engineering Mathematics | Microelectronics | Digital Signals and Filters | Linear Algebra and Differential Equations

EXPERIENCE

Student Makerspace Lead, Chapman University	September 2022 - Present
- Supported student engineering projects from concept to fabrication, ensuring timelines and safety procedures - Developed and maintained comprehensive, streamlined instructions for equipment operation and sample project implementation - Organized and delivered hands-on technical workshops to train students on fabrication tools, electrical assembly, and proper safety - Operated in a laboratory fabrication environment applying proper PPE practices, lab safety standards, and equipment operating protocols.	
Electromagnetics II Teacher's Assistant, Chapman University	August 2024 - December 2024
- Modeled antennas using Ansys HFSS software for advanced electromagnetic field simulations - Developed curriculum support materials to assist the professor in enhancing student learning	
Introduction to PCB Design and Fabrication Teacher's Assistant, Chapman University	August 2025 - December 2025
Guided students through troubleshooting PCB/PCBA design and fabrication on LDKF tools	

ELECTRICAL DESIGN & HARDWARE PROJECTS

Target Retrieval System for Gun Ranges (Independent Contracting)	December 2025-Present
- Designed and fabricated an industrial PLC-controlled motor system with touchscreen HMI and three-phase motor control - Programmed PLC logic for real-time motor control, sequencing, and developed HMI interface with serial communication between the HMI and PLC - Designed and tested custom PCBA and power distribution, including signal conditioning, voltage regulation, and multi-voltage rails from 110V AC using lab instrumentation (oscilloscopes, multimeters, power supplies) - Managed PCB fabrication workflow by preparing design files, selecting components, and coordinating with external manufacturers for production	
Life-Sized Functional R2-D2 Replica from Star Wars (Robotics Club)	February-August 2025
- Led development of a life-sized remote-controlled R2-D2 robot as project lead across mechanical, electrical, and software teams - Designed custom PCB lighting systems and integrated control electronics for movement, sound, and user interface - Built a Raspberry Pi-based controller and networking interface to operate the robot remotely	

ACHIEVEMENTS & ACTIVITIES

Chapman Robotics Club, Secretary	Fall 2024 - Present
Society of Manufacturing Engineers, Secretary	Spring 2025 - Present
Fowler School of Engineering Deans Scholarship, Recipient	Fall 2022 - Present
Chapman University Provost List, Member	Spring 2023, Spring 2024, Fall 2024, Spring 2025
Chapman University Computer Science Club, Member	Fall 2022 - Present